

Open-RoadAtlas: Leveraging VLMs for Road Condition Survey with Real-Time Mobile Auditing

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ABSTRACT

Road surveying plays a vital role in effective road network management for local governments. However, current practices pose challenges due to their costly, time-consuming, and inaccurate nature. In this paper, we propose an automated survey platform that supports weed, defect and asset monitoring with instance segmentation models. Empowered by recent advancements in vision-language models (VLMs), our solution offers improved flexibility for novel tasks with a limited label set. For domain specific classes, such as pavement cracks and potholes, we train a detector to identify their location given our sparsely annotated images, and alleviate false-positives by rejecting predictions outside regions of interest identified by VLMs. The proposed system directly involves managers in the survey process through a mobile application. The application allows three core functions: 1) capture and cloud upload, 2) real-time survey trajectory monitoring, 3) open-vocabulary detection.

CCS CONCEPTS

• Computing methodologies → Object detection; • Computer systems organization → Embedded systems.

KEYWORDS

road asset management; defect detection

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1 INTRODUCTION

Road surveying is an essential task for local governments to effectively manage their road network. However, current practices often hinder effective management, as they require costly, time-inefficient, and inaccurate methods. The majority of road agencies

perform surveying using specialised machines such as laser profile meters, or by hiring experienced inspectors, collecting defect information by hand. These procedures are both prohibitively expensive (\$350–400k per survey), and prevent local government from conducting frequent surveys to keep data up to date.

Developing an automatic defect detection system using machine learning can be a complex task [2]. Local governments need to invest a significant amount of time in planning data capture and annotation tasks, which can take years of continuous data integration and system development. As a result, managers are often hesitant to develop a system that involves training a network with supervised learning for a new task. However, recent research advancements have led to the creation of foundation models with satisfactory zero-shot performance for novel tasks. Additionally, efficient fine-tuning techniques like prompt learning and prompt engineering now make it cost-effective and efficient to adapt large models to new downstream tasks.

We present a system framework that enables road asset management using domain-specific models alongside vision-language models. The framework allows for the management of new tasks with specific requirements whilst minimising the required number of labelled data points. We utilise rules-based filtering and semantic relations to remove semantically incompatible predictions from being propagated to our asset database. The framework allows for seamless monitoring through a bespoke mobile application.

2 SYSTEM FRAMEWORK

Our system¹ is comprised of two main components. The first is the backend, where we store all our defect data, and offer it through a Flask API. This serves as our defect information for our mobile application and web dashboard. It is also responsible for processing the incoming images. We use multiple task-specific models, as well as a vision-language model, to filter out semantically erroneous predictions. Secondly, we have our mobile application, which can be used by road managers to inspect surveys on-site, and capture model predictions through our camera interface.

2.1 Backend

The system backend facilitates the processing of images with large deep-learning models, and the storage and querying of defect data for our mobile application. We utilise Grounding DINO [4] to detect boxes for text prompts, and then feed the boxes to Segment Anything [3] to produce instance masks. We have a number of in-house

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¹Technical Demonstration Video

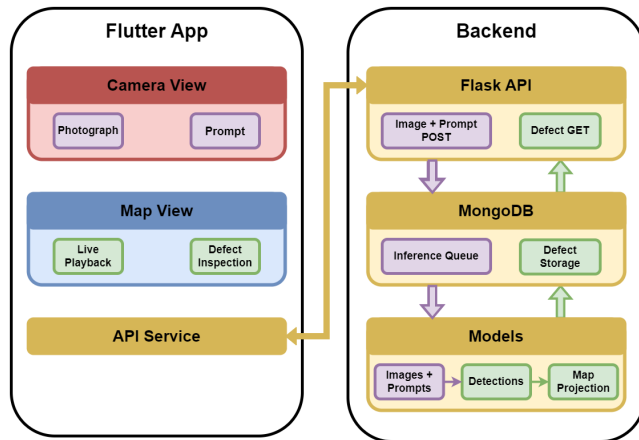


Figure 1: System Framework

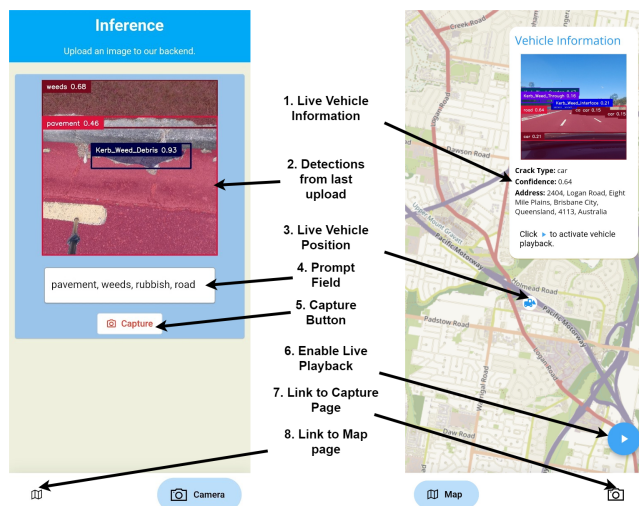


Figure 2: App Diagram

models for defect detection, leveraging Cascade RCNN [1]. We keep an inference queue of images, their GPS coordinates and prompts in our MongoDB database, then infer them in a streaming order, projecting them to map view with the given GPS coordinates. The backend reduces the number of false-positives by filtering detected boxes that overlap with negative classes prompted into the open-vocabulary model. The open-vocabulary model benefits from large scale training, offering strong generalisation performance.

2.2 Mobile Application

The mobile application has been developed with a focus on three core functions: capturing snaps and uploading them to our cloud server to update our defect database, visualising survey vehicle trajectories and gathering road condition statistics, and providing open-vocabulary detection for seamless addition of new classes.

The two major views of the application are shown in Figure 2. The left screenshot shows the capture page which is shown when

the app is first loaded. The prompt field (item 4) offers the user the ability to enter a prompt for the open-vocabulary model. They can then capture (item 5), the app will then upload to our cloud server and present the uploaded image with instance masks (item 2). On the right hand side, we have the map page. The map page allows real-time survey monitoring and detailed inspection. It offers a live playback function (item 6) which enables the live detections to be displayed as map markers, and the latest information is shown in a card widget in the top right (item 1).

2.2.1 Capture and Cloud Upload. The capturing page of the app enables users to interact with the defect detection system through a low latency means, enabling them to seamlessly interact with the system and gain confidence in its performance. This also functions to update our defect database, enabling accurate and up-to-date information.

2.2.2 Real-Time Survey Trajectory Monitoring. The app enables the real-time management of active survey vehicles, to visualise their trajectories and plan the best course. It offers statistical insights into the condition of roads, and generates informative reports and metrics that assist in assessing road conditions. This functionality enables efficient monitoring and decision-making for road condition projects.

2.2.3 Open-Vocabulary Detection. Lastly, we incorporate an open-vocabulary detection [5] model. This allows users to add new classes or categories to the application's detection capabilities. By utilising vision-language models, the application can be fine tuned to recognise and identify defects or assets beyond the existing pre-defined classes. This allows managers to adapt the application to their specific needs and expand its capabilities over time.

3 CONCLUSION

In this technical demo, we have presented a novel system for seamless road defect management. The system combines the convenience of capture and cloud upload, advanced visualisation of survey vehicle trajectories, statistical analysis of road conditions, and the flexibility of open-vocabulary detection. By incorporating these three functions, the application aims to enhance defect management, facilitate informed decision-making, and offer adaptability to accommodate evolving requirements.

REFERENCES

- [1] Zhaowei Cai and Nuno Vasconcelos. 2019. Cascade R-CNN: High Quality Object Detection and Instance Segmentation. arXiv:1906.09756 [cs.CV]
- [2] Zhuoxiao Chen, Yiyun Zhang, Yadan Luo, Zijian Wang, Jinjiang Zhong, and Anthony Southon. 2021. RoadAtlas: Intelligent Platform for Automated Road Defect Detection and Asset Management. In *MMAasia '21: ACM Multimedia Asia, Gold Coast, Australia, December 1 - 3, 2021*, Changwen Chen, Helen Huang, Jun Zhou, Tatsuya Harada, Jianfei Cai, Wu Liu, and Dong Xu (Eds.). ACM, 62:1–62:3. <https://doi.org/10.1145/3469877.3493589>
- [3] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. 2023. Segment Anything. arXiv:2304.02643 [cs.CV]
- [4] Shilong Liu, ZhaoYang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. 2023. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. arXiv:2303.05499 [cs.CV]
- [5] Jianzong Wu, Xiangtai Li, Shilin Xu, Haobo Yuan, Henghui Ding, Yibo Yang, Xia Li, Jiangning Zhang, Yunhai Tong, Xudong Jiang, Bernard Ghanem, and Dacheng Tao. 2023. Towards Open Vocabulary Learning: A Survey. arXiv:2306.15880 [cs.CV]